Using Artificial Intelligence to Augment a Social Robot That Supports Situationally Impaired Participants in Hybrid Meetings

Abstract

The growth in hybrid working arrangements has resulted in the normalisation and acceptance of videoconferencing in place of traditional meetings. When meetings occur in which some participants are physically present and others participate remotely, there is a tendency for the former group to be conversationally dominant over the latter. This paper explores how artificial intelligence can be used to enhance a robotic surrogate, enabling remote participants to more fully participate in meetings, especially when they are unable to use a camera or microphone.

In a prototype of the concept, a social robot is integrated with video-conferencing software and several AI agents. The use of artificial intelligence is critical as it reduces the cognitive burden of operating the robot, allowing the remote participant to devote their attention to the meeting content. A pilot study was undertaken with five participants, the findings are presented in this report and are used to inform several recommendations for future extensions of the concept of using a social robot augmented by artificial intelligence for use in hybrid meetings.

Keywords: Artificial Intelligence, Hybrid Meetings, Situational Impairment, Social Robots, User Experience

1 Introduction

The COVID-19 pandemic drastically accelerated a continuing trend towards remote working [24]. This represents one of the largest shifts in how people work in recent decades. Hybrid meetings (with both co-located and remote participants) are becoming an increasingly important part of many work environments.

The use of hybrid meetings offers flexibility (and cost savings) to those unable to attend in person [33]. Despite these clear advantages, it does create several problems, one is *in-room dominance*, co-located participants often tend to dominate the conversation [37]. This problem is magnified when the remote participant is *situationally impaired*, such as being unable to use their camera or microphone due to technical issues, privacy issues (if participant is a child) or other circumstances (travelling on public transport). The remote participants can often feel sidelined and unable to properly participate in the meeting [8].

One innovative technique to assist remote participants is the use of mobile teleoperated robots, serving as their avatars [4]. These robots act on behalf of the remote participant and can lessen the degree to which they are excluded from a meeting. However, the cognitive burden of operating the robot reduces the amount of attention that the remote participant can devote to following the meeting. For situationally impaired participants, these robots are especially challenging to operate, as participants have to communicate via typed answers, introducing a degree of latency in their responses and causing them to feel frustrated [48].

The last decade has seen artificial intelligence technology make major advancements, driven by changes in hardware, the creation of new algorithms and the massive amounts of data created through the internet [41]. The applications for artificial intelligence are widespread [22, 46], including significant opportunities to improve human-robot interaction (HRI) [21, 28].

This report proposes the use of artificial intelligence to help control a robot proxy (with speaking, gestures etc.), enabling it to be used in hybrid meetings to mitigate the problems faced by remote participants, while also improving the experience of co-located participants.

After some analysis on the existing literature around telepresence and embodiment, situational impairment in hybrid meetings and artificial intelligence for social shared control, a basic prototype is presented for an AI-augmented robot proxy system. The system was evaluated in a brief pilot program, where participants had a chance to review the capabilities of the system and provide feedback on its performance.

The results of this pilot study indicate that the use of artificial intelligence to assist with controlling a social robot has the potential to be transformative, allowing a social robot to be operated by a remote participant without causing undue stress. The results also inform several suggested improvements that would greatly improve future iterations of the prototype, including the reduction in conversational latency by running the entire system locally, using audio in addition to video data to determine who is speaking, and improving the quality of suggested responses.

2 Background and Related Work

This research draws upon existing work done on situational impairment in hybrid meetings, telepresence and embodiment, as well as artificial intelligence for social shared control.

2.1 Situational Impairment in Hybrid Meetings

When contextual factors affect a user's interaction with a device, these are referred to as Situationally Induced Impairments and Disabilities (SIIDs) [15]. Existing research has focused on considering situational impairment through the same lens as Health Induced Impairments and Disabilities (HIIDs) [39]. Another area frequently explored is when situational impairments affect the use of mobile devices [38], especially due to divided attention, loud noises or the weather [47].

Regarding the problems facing remote participants in hybrid meetings, some novel ideas have been proposed to create more equitable and inclusive hybrid meetings - such as the Hybridge system, in which co-located participants interact with remote participants that are represented by displays around a table and the remote users are provided a 3D view of the meeting [31]. Other research investigated the use of a "virtual room" in which every participant's

background is removed and everyone is placed in a common scene [43].

While these solutions have potential, very little work has been done regarding conditions where remote participants are unable to use their microphone or camera to participate, or cases in which situationally impaired participants control a telepresence robot.

This study begins to address this shortfall, by proposing an AI-augmented social robot system that can be easily used by a situationally impaired remote participant.

2.2 Telepresence and Embodiment

Since the term "telepresence" was originally coined in 1980 [25], a great deal of work has been done to increase the immersive experience of remote users, with telepresence robots being utilised in education [20], healthcare services [18] and work environments [19]. Current telepresence robots can provide unique experiences and opportunities for remote users, allowing them to communicate through non-verbal cues and use the robot's locomotive movement [40]. However, most commercially available telepresence robots use a common decades-old design (see Figure 1), commonly referred to as "iPad on a stick".

Figure 1: Examples of traditional telepresence robots - The Beam family of robots: Beam, BeamPro, and BeamPro 2. Photo: Suitable Technologies [44]

Compared to "humanoid" robots, these traditional telepresence robots limit the degree of embodiment for the user [16], moreover, these designs do not address cases when remote participants are situationally impaired.

In the context of telepresence, embodiment refers to the extent to which a remote user in a different location can experience a physical presence [3, 45]. A higher degree of embodiment increases the remote participant's connectedness, with results showing that systems with a higher degree of embodiment increase the levels of communication between participants [7]. Empirical research also shows that for "humanoid" robots, participants feel that replacing a robotic face for a human face increases embodiment [50]. This study focuses on embodiment using the Furhat robot system (see Figure 2), which has a humanoid head that can project different human faces.

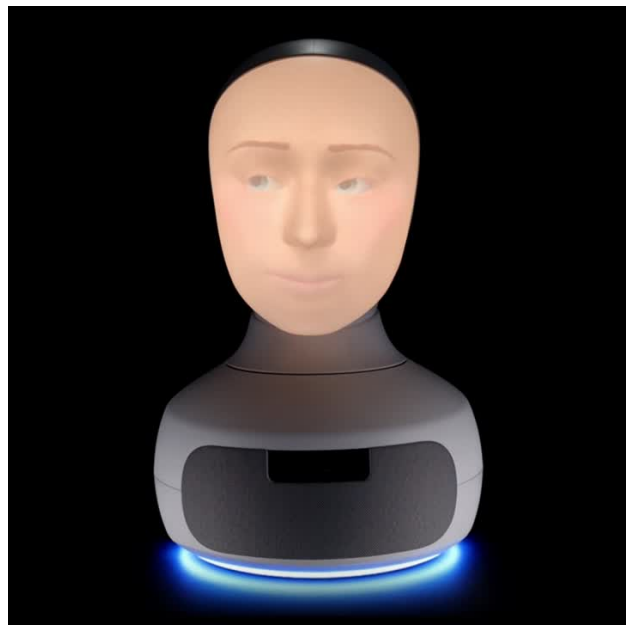

Figure 2: Furhat with LED light in robot base turned on [35]

While telepresence robots can be beneficial for remote participants, they can present several issues, many co-located users find the psychological experience of interacting with a robot possessing human-like features to be challenging [2], other users have privacy concerns when interacting with these robots [27] or simply have a strong predisposition for face-to-face interactions. These issues demonstrate that the challenge of seamlessly integrating social robots into hybrid meetings remains ongoing. This integration (of social robots) is crucial to enable a high

degree of embodiment for remote participants and to minimise the communicative and social impediments that they experience.

2.3 Artificial Intelligence for Social Shared Control

Recent improvements in artificial intelligence have transformed the way that humans interact with robots, with much greater potential to foster effective and meaningful interactions [23]. In terms of research that addresses how humans use and interact with robots, a great deal of research is focused on ethical issues [12, 29]. There is also a great deal of research on using AI to augment robots that perform certain tasks for a person [14], this can be referred to as *assisted intelligence* [5, 13].

In comparison to assisted intelligence, work is also being done on *augmented intelligence*, where a person's own abilities are boosted by AI, tools such as large language models (LLMs) [26] and AI systems that can identify points of interest in medical imaging data [9, 32] fall under this category.

Regarding augmented intelligence, some work has been done in relation to social shared control, but this largely involves tasks such as using LLMs to help robots learn how to perform certain tasks faster [49], there is limited work in terms of using artificial intelligence to augment a robot acting on behalf of a remote participant in hybrid meetings. Regarding the use of artificial intelligence to empower remote participants, most work done has been limited to using software, such as using AI to help remote participants take notes or performing sentiment analysis to inform the remote participant on how the co-located participants feel [6, 30], there has been little work on using AI to empower a remote user by assisting them to operate a robot.

This study aims to address the deficit, by investigating the use of artificial intelligence to assist with controlling a social robot during hybrid meetings.

3 Prototype System Design

The AI-enhanced telepresence system designed for this study consists of the co-located Furhat robot, a web interface, two webcams (one with an in-built microphone), and several AI systems. The Furhat robot functions as a representation of the remote participant, they have some control over the robot's actions (through the web interface), with varying degrees of control ceded to artificial intelligence agents which infer the remote participant's intent. Initially, code was provided from a previous study, but this code had a basic user interface, was not compatible with mobile devices and lacked the AI-related features mentioned in this section of the report.

3.1 Improved UI, Tablet Compatibility, File management and Documentation

While not directly related to the functioning of the system, major work was undertaken to update the user interface to ensure it was easy to use. This included the use of contrasting colours, responsive buttons that changed colour when highlighted and use of on-screen pop-ups for certain functionalities. The user interface was also made compatible for tablets. A meeting file management system was implemented, allowing meeting transcripts to be saved and then reloaded for future use. For the continuity of the project, extensive documentation was created outlining key aspects of the code and installation instructions, the updated code had extensive commenting added, supporting code extensibility.

3.2 Suggested Responses

The existing telepresence system for the Furhat robot required all spoken answers to be typed by the participant. The process of typing out responses from scratch is both time-consuming and distracting for the remote participant. It also renders them at a relative disadvantage to co-located participants who can quickly speak their reply.

As a remedy, this prototype uses an AI agent (from OpenAI) to assist with drafting responses, which will reduce conversational latency. The model used for the AI agent was GPT-4o-mini, this model is one of the fastest models currently available from OpenAI [42]. To ensure that answers are appropriate, the agent is provided with two contextual sources of information.

The first is the initial context regarding the meeting (around 50 words) which is provided by the remote participant prior to the meeting's commencement. The second source is a tran-

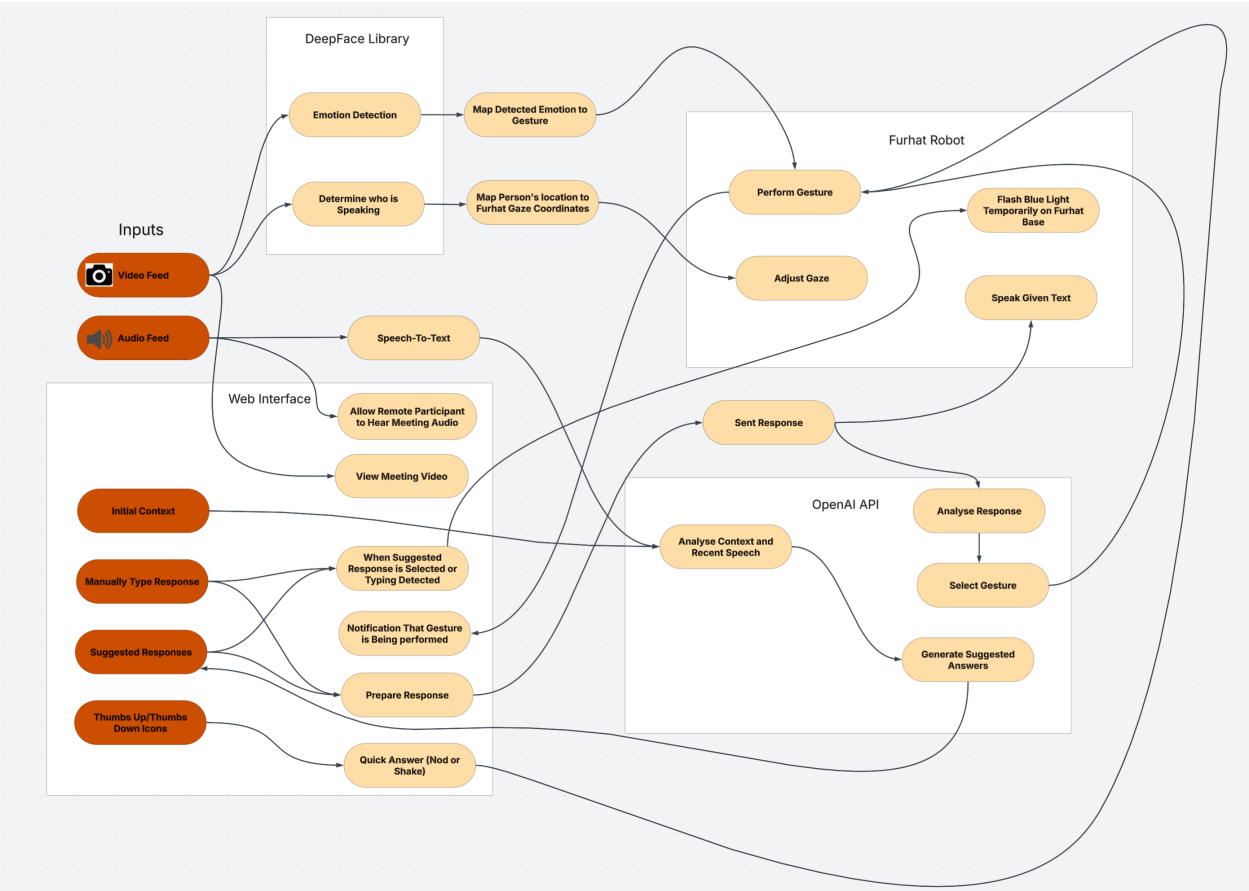

Figure 3: Outline of overall workflow of the system during the pilot study

script of the meeting which is generated using Whisper (OpenAI's open-source, speech-to-text model, run locally on a laptop) [34]. Whisper has been released in several sizes, the small version is used for the pilot study. The AI agent only considers the previous few minutes of the meeting when formulating responses.

Three possible responses are provided, they can be used immediately or edited by the remote participant. The system still allows the remote participant the opportunity to type their answer from scratch if they are not happy with any of the responses. The suggested responses are updated approximately every 10-12 seconds. Whenever a participant is typing an answer or they select a suggested response, the base of the Furhat will glow blue to inform co-located participants that the remote participant wishes to say something.

3.3 Quick Answers

To improve conversational latency, two permanently available quick answers are provided ('yes' and 'no') Clicking one of these two buttons will cause the Furhat robot to quickly respond with a nod or a shake of its head.

3.4 Automatic Gestures

The Furhat robot can perform a variety of gestures (Smiling, Nod, Frowning etc.). The existing system provided only allowed for five preset gestures to be performed, they also had to be triggered manually by the remote participant. This limited the extent to which the robot could be used for non-verbal communication.

The solution is to use a second AI agent that analyses the lines of dialogue typed by the remote participant (and then spoken by the robot). Based on what the Furhat is saying, the AI agent will look at the full list of gestures that it can perform and automatically select the most optimal one. This reduces the cognitive

burden placed on the remote participant. By default, the Furhat can perform nineteen different gestures (see Appendix B).

A second variant (not used during the pilot) of this functionality is rather than automatically performing gestures, the AI agent will merely suggest 2-3 appropriate gestures based on what is being said and allow the remote participant to decide which one to execute (see Appendix A2).

Whenever any gestures are triggered (either manually or automatically), the participant is notified on the user interface.

3.5 Active speaker and emotion detection

The Furhat robot has basic in-built functionality to detect which co-located participant is speaking. Unfortunately, it is solely based on proximity, which limits its usefulness in a meeting.

For this study, a system using Python's DeepFace library [17] has been developed which takes a camera feed as input to determine the location of the co-located participant who is speaking. The system identifies the location of every face visible to the camera and then infers who the active speaker is based on the person who has opened their mouth the most in the preceding few seconds. The Furhat robot will then turn and shift its gaze towards that person. Until a new speaker 'takes over', the system will continue paying attention to the co-located participant who spoke last.

Initial variants of this system were highly responsive (constantly shifting gaze), this was distracting and detracted from the overall experience of interacting with the system. The version presented for this study will only shift its gaze after at least 7-8 seconds.

The same system also infers the emotion of the participant who is currently speaking, based on the emotion inferred, on occasion, the Furhat robot will reciprocate an emotion (e.g. smile back at someone who is smiling).

4 Pilot Study

Due to the limited time available, a full and thorough survey was not undertaken. Instead, a pilot study was performed to gain a basic understanding on how easy the system was to use and whether participants had any further suggestions for improvement. All 5 participants were members of the Human-Computer Interaction Group at the University of Melbourne, a clear advantage to this was that all participants were able to give highly relevant feedback on the system. One potential issue is the lack of diversity among the participants [11], best practice for future studies will involve recruiting participants from a diverse set of backgrounds.

4.1 Pilot Design

The pilot was broken up into three sections, orientation, main task and answering questions. The orientation simply involved explaining the purpose of the system and providing a basic outline of what the participant's role in the pilot study would entail.

For the main task, the participant would spend time acting both as a co-located and a remote participant.

4.1.1 Setup:

As a co-located participant, they faced the Furhat robot (see Figure 4), with a second camera present to capture the video feed for the Zoom meeting. When acting as a remote participant, they were located separately from the robot. An iPad mini was used to control the system.

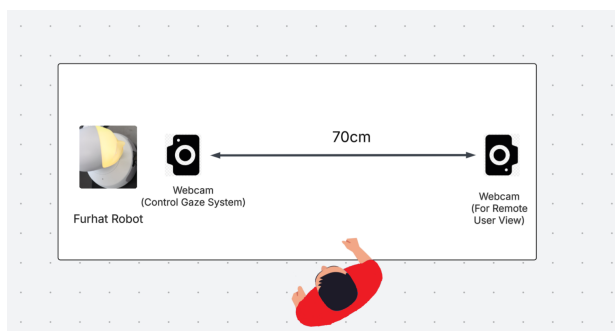

Figure 4: Diagram of co-located setup

4.1.2 Main Task:

First they would act as the remote participant, with the following tasks being performed:

- Send a message in response to a simple question (e.g. how are you today?)
- Respond to a more complex question or make a statement about a chosen topic

- Attempt to interject while a co-located participant is speaking
- Have a brief conversation with the demonstrator (who is playing the role of a co-located participant).

These tasks were performed twice, with the artificial intelligence system being disabled the first time (see Appendix A1), allowing for the participant to compare the amount of effort required to operate the system (with and without AI augmentation).

They then acted as a co-located participant, testing out the active speaker detection system by changing where they were sitting relative to the Furhat robot, they also spent time deliberately smiling, frowning etc. (to gauge the emotion detection system), they then participated in a short discussion with the demonstrator (playing the role of a remote participant) to experience the system properly as a co-located participant.

Throughout the demonstration, the participant was encouraged to "think aloud" sharing their thoughts and feelings about their interactions with the system [36].

4.2 Post Study Questions

After the participant had concluded interacting with the system, they were invited to answer some qualitative questions (see Table 1).

Questions for Using the AI-Enhanced Telepresence System
1. How would you feel about using the Furhat Robot (in general) in a hybrid meeting setting?
2. As a remote participant, did the artificial intelligence augmentation improve your experience using the system?
3. As a remote participant, were the suggested responses broadly appropriate?
4. As a co-located participant, did interacting with the Furhat robot help increase your sense of connectedness to the remote participant?
5. Do you have any suggestions to improve the system or the meeting experience?
6. Do you have anything else you would like to add?

Table 1: Survey Questions for the Pilot Study

4.3 Pilot Study Findings

All quotes provided by participants (with their consent) are anonymised with only an ID being provided (e.g. P1). Overall, participants gave positive feedback of the system in general, being receptive to the concept of using a robot as a proxy and for the use of artificial intelligence to help them better control the robot. Comments such as, *"In some conditions where we cannot directly speak, I think it's a huge improvement than just typing, with the help of the Furhat's voice, facial expressions and (head) movement"* (P2) and *"I think it was cool to use a robot for a meeting as a replacement of a human in the situations where the person cannot be present"* (P3) were typical.

When the AI enhancements were disabled, participants often found it challenging to keep up with the conversation due to the time it took to type out an answer.

This was largely in line with expectations, that the benefit of having a social robot act as an avatar for a remote participant was reduced due to the amount of work required to properly operate it.

When the AI augmentation was activated, participants generally found the suggested responses to be broadly relevant to the topics being discussed. However, there was some variation in the quality. Whenever the topic was fairly general (e.g. talking about the future of AI), participants found the suggested responses to be quite useful with comments such as, *"Broadly, the suggestions were close to the message I wanted to convey"* (P1). However, whenever the topic became much deeper, such as discussing the use of reinforcement learning to improve social robots, the system often struggled to give specific details, with suggested responses remaining generalised, with one participant commenting, *"It would pick up the key words of our discussion, but left the topics (of the discussion) at a high level"* (P5). Another limitation was the system's difficulty in handling a change in topic, it would require a minute or two of back-and-forth between the

participant and the demonstrator for the system to suggest responses from the new topic, this issue was encountered by most of the participants, with one commenting, *"I attempted to change the conversation to see if the LLM would update it (the responses), but the system focussed on the previous topic"* (P5). Some of the problems with the suggested responses could be attributed to the fact that due to limited computing power, the version of Whisper used was one of the weaker models, resulting in more errors when transcribing the meeting.

Several interesting recommendations were provided by the participants. One suggestion mentioned by several participants was to cease the automatic generation of suggested responses and allow the participant to decide when they were generated (e.g. such as clicking a button). Another suggestion made by a few participants was for the addition of some type of tonal control (i.e. the remote participant chooses in which tone the robot responds), with participants making comments such as, *"Adding a feature to add tones to the text (spoken) could also be interesting"* (P1).

One suggestion was to ensure that the three suggested responses should be as different from each other as possible (this participant reported that the suggestions were all too similar). Another recommended a system in which the remote participant selects one of the suggested responses which will cause the system to regenerate three responses which are variations of the response selected but with more detail - effectively allowing the remote participant to fine-tune their answer.

Another interesting finding was that participants who were faster at typing reported less use of the suggested responses compared to those who were slower at typing. When asked, the participants who typed slower indicated frustration or stress at being unable to respond quickly - they felt that a loss in accuracy by resorting to the suggested responses was more than made up by the increase in speed. For those who could type faster, the relative increase in speed by using suggested responses was much less, with one participant stating, *"I preferred using the system manually (not using suggested responses), that is likely because I am a reasonably fast typer and care about my opinion being conveyed as accurately as possible"* (P5).

Regarding gestures, when the AI augmentation was switched off, most participants were too engrossed with typing out their responses to also properly respond with gestures. Most preferred to have the automatic gesture feature enabled, as they felt that it had the potential to add non-verbal communication to their selected answers. However, one participant indicated a preference for manually controlling all gestures, *"When I was manually controlling the gestures, I could add a wink or a head tilt to my comment at an appropriate moment in order to change the subtext of what was said"* (P5). For gestures, most participants indicated that the delay (due to running the gesture system via API) affected the immersion of this feature, there was also a desire for the range of gestures chosen to be wider, as the AI agent would usually default to choosing the same few gestures often (Nod, Smile, Surprise, BigSmile, Think, Frown).

When acting as the co-located participant, everyone had positive opinions about the active speaker detection feature, liking the idea that the Furhat is "paying attention" as if it was a real person, with typical comments such as, *"I enjoyed having a humanoid face to view, and the Furhat gave me a point of conversation with the remote participant"* (P5) and *"I felt that the person controlling the robot was paying attention to me"* (P4). The main recommendation regarding this feature was to also consider the use of sound in tandem with detecting when someone opens their mouth as the current system sometimes struggled to detect when a participant was talking, especially when they were speaking quickly. Some of the quality issues with the active speaker detection were caused by the limited computing power available, as the system could not run at maximum settings without crashing.

There were some comments around the latency, in the sense that the co-located participant had to wait a few seconds in between answers, with participants commenting, *"In the future it (the robot system) might increase the sense of connectedness, but for now the responses are not very fast"* (P3), *"It is a really cool feature, but the delay (generating suggestions) makes it hard to use"* (P4) and *"The responses showed a bit of a delay in the con-*

versation, and the time (to generate responses) should be reduced" (P2).

In terms of technical difficulties, there were some minor connectivity issues between the web interface and the Furhat robot, these usually resolved themselves quite quickly. For one participant, there was a connectivity issue with the API, which required the system to be quickly reset. A few times, the webcam used for the remote participant view stopped working, so the webcam used to control the gaze system was also used for the Zoom meeting. Overall, these technical difficulties didn't materially affect the overall participant experience. Given the background of the study participants, most were quite understanding regarding the technical issues and did not adversely change their opinions on the system due to them.

5 Limitations

This study was limited by several factors. Due to the short development timescale, a thorough survey was not possible. It would be highly beneficial for a thorough study to take place and a short proposal for a full study is presented in the next section. As highlighted in the results, the quality of participant interactions with the system during the pilot was affected by the limited amount of computing power available. Additionally, the use of agentic AI systems accessed via API introduced some latency to the functioning of the system and occasionally impacted the system's reliability.

6 Future Extensions

This current project could be extended in several ways, both in terms of additional evaluation of the system presented and future improvements to the system.

6.1 Full Survey

A full qualitative and quantitative study could be completed to objectively measure the usefulness of using AI to augment the existing Furhat system. This study would involve six groups of three people each (18 total participants), each group would have two people acting as 'co-located' participants, while the third person would function as the remote participant. Six groups is within the recommended sample size when conducting focus groups for evaluating a product [1]. For participants to evaluate the system without any bias, the participants will only have a single role (i.e. remote participants will not then have a chance at acting as a co-located participant).

The participants would spend 45 minutes interacting with the system. After an initial briefing, the participants would complete a set of collaborative activities, simulating real-world meetings which would allow them to thoroughly evaluate the system from the perspective of a remote and co-located participant.

For the full survey, it is essential that participants are drawn from a variety of backgrounds, especially those who are not from an engineering or computer science background, their opinions will be crucial to ensure that the system can be useful to as many people as possible.

6.2 Run all AI Locally

Future iterations of this system would greatly benefit by running all artificial intelligence functions locally and on a computer with more processing power. This would ensure that both the speech-to-text model and the active speaker detection feature would both perform much better. Generating the suggested responses locally would significantly reduce the latency of the system - leading to higher user satisfaction and an improvement in reliability, by avoiding API connectivity issues. For the automatic gesture system, this would allow gestures to be triggered almost concurrently to when the robot begins speaking - improving the immersion of the system. While running all AI functionality locally would certainly improve the system, it should be noted that it could be quite expensive to acquire the necessary hardware as modern LLMs have billions of parameters (requiring large amounts of compute).

6.3 Improving the Quality of Suggested Responses

Out of all the issues identified by participants, the quality of suggested responses was cited by most as the crucial feature to make improvements on. Improving the quality of the suggested responses is a complex task, two potential strategies to improve the quality of responses are presented here.

6.3.1 Speaker Recognition

A future extension to the existing system could incorporate speaker recognition, instead

of merely transcribing the meeting and then providing suggested responses, a new system could use voice recognition technologies to determine the identity of which co-located participant is speaking, allowing for the suggested responses to be tailored towards each co-located participant. Moreover, this could be used to improve the active speaker detection feature - giving the system an additional indicator on who is speaking.

6.3.2 Improved Context System Using Retrieval-Augmented Generation

Rather than simply allowing the remote participant to type some basic information about the upcoming meeting, a future system could be much more complex. Many LLMs perform well on generalised information, but often struggle with precise topics that were largely absent from their training data. Using Retrieval-Augmented Generation (RAG) offers the possibility of using data from sources outside of a large language model's own knowledge [10]. These sources could involve documentation about the issue being discussed, summaries or transcripts of past meetings and even some biographical information about each of the meeting participants.

6.4 Fully Automated Responses

Some initial work was done to develop a fully automated system, that could interject and ask questions on behalf of the remote participant, the system was not used in the pilot as it would constantly interject, even when someone was speaking. A future extension to this current study would involve further work on this idea, a proposed agentic workflow is shown (see Figure 5), a key hurdle would be ensuring that the system will only interject at appropriate times.

7 Conclusion

In aggregate, the AI-enhanced robot telepresence system presented demonstrates high potential for reducing the situational impairment experienced by remote participants in hybrid meetings while also improving the experience of the co-located participants. This report supplies an innovative outlook into the integration of artificial intelligence with social robots to reduce situational impairment in hybrid meetings for remote participants. Through use of artificial intelligence, remote participants can control a robot which physically embodies them in the meeting room without imposing a high cognitive burden. The short pilot study performed supports the feasibility of integrating AI with the Furhat system, with good overall performance and general feedback being positive, however, before the system can be used in real-world situations, it will be necessary to improve the latency of AI-suggested responses and the quality of several other parts of the system. Future studies could focus on implementing some of the recommendations presented in this report to improve the current system in addition to conducting a more thorough quantitative study.

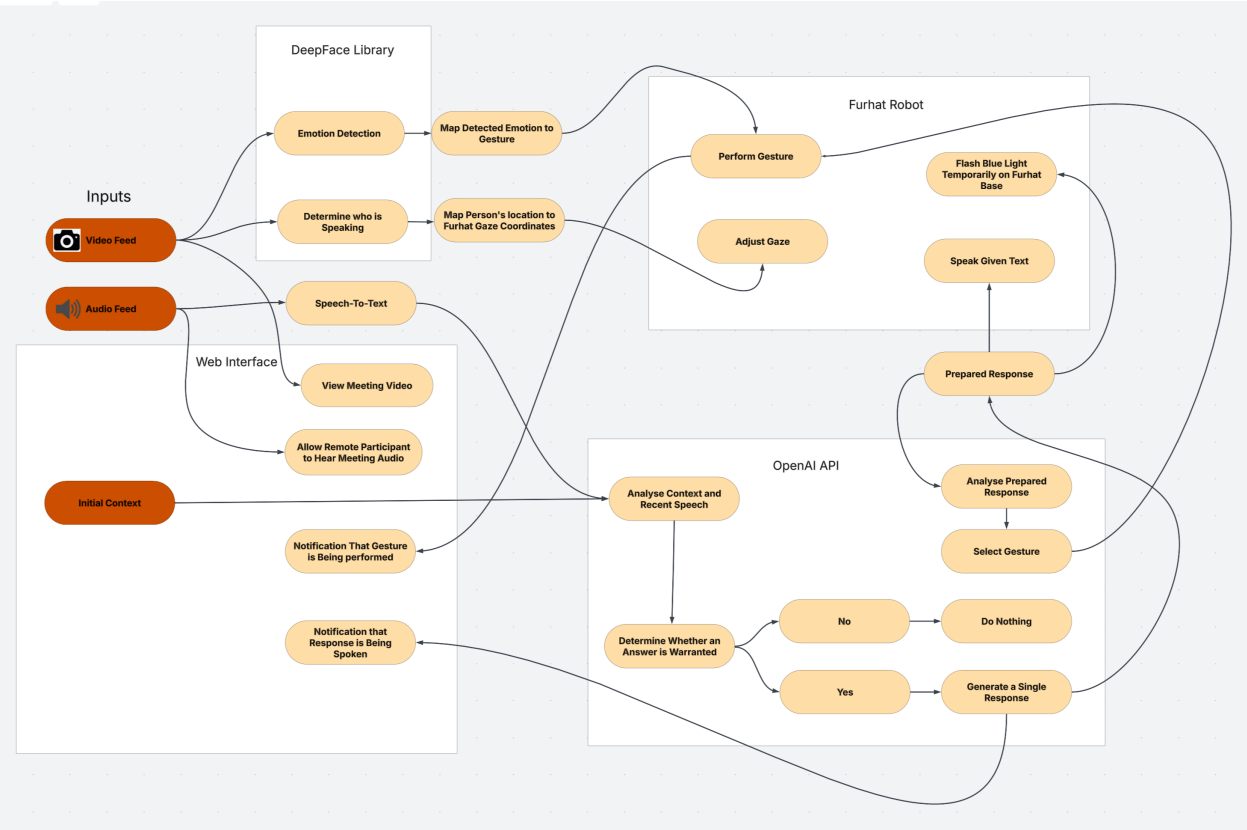

Figure 5: Example of workflow for fully automated system

8 Appendix A: Additional Diagrams

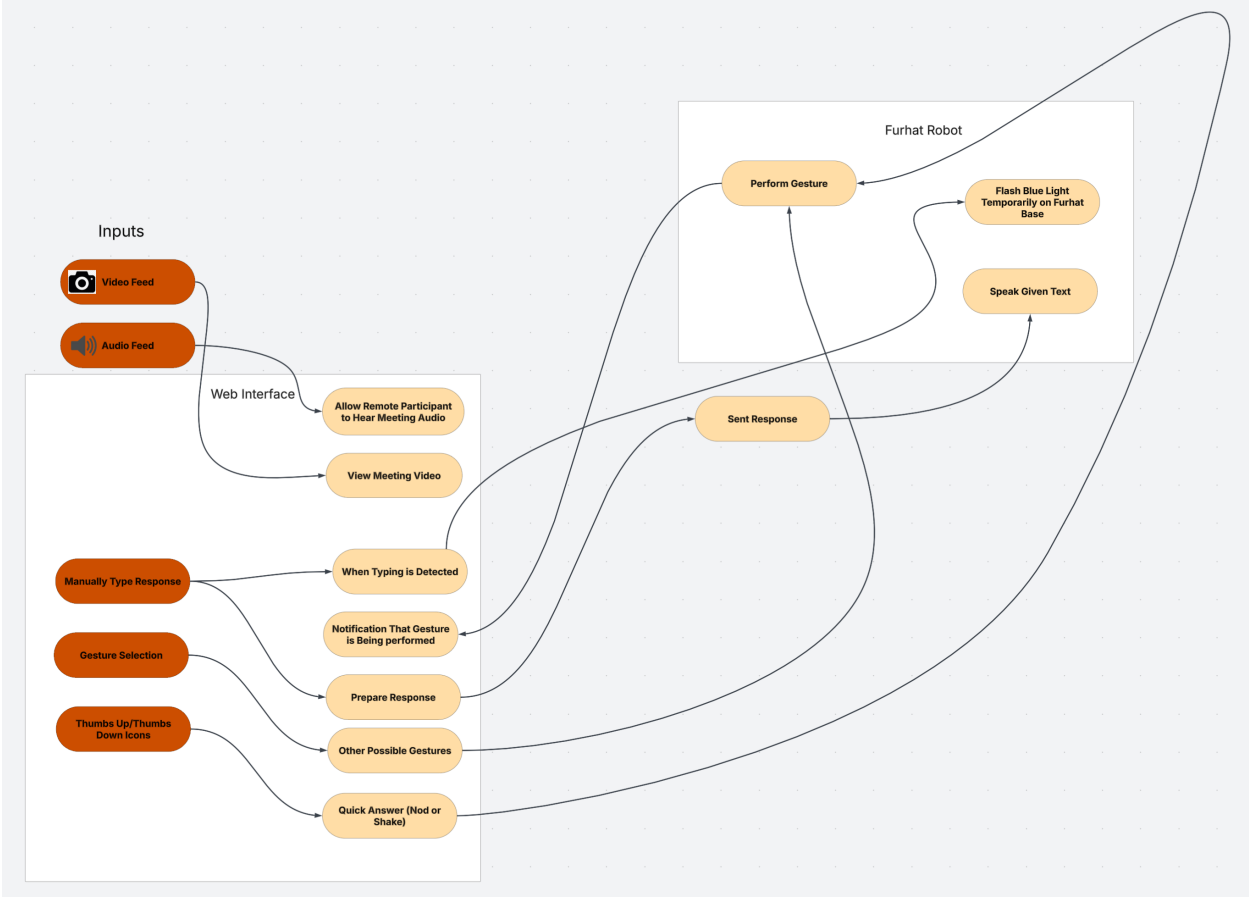

Figure A1: Outline of the workflow of the system with all AI disabled

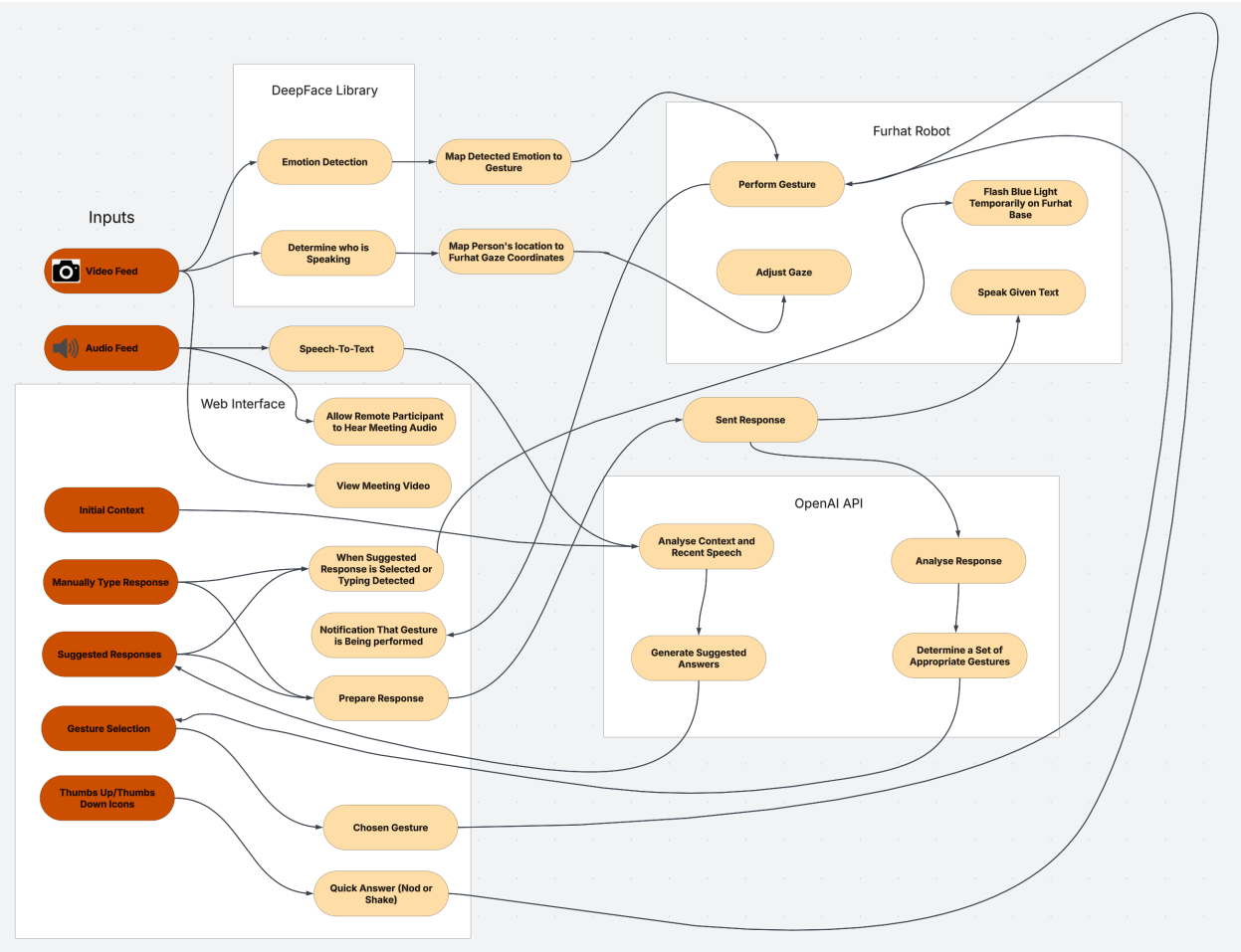

Figure A2: Outline of the workflow of the system with suggested responses and suggested gestures (i.e., system recommends some gestures)

9 Appendix B: Full List of Default Furhat Gestures

- Nod
- Shake
- Smile
- Frown
- Surprise
- RaiseEyebrows
- Blink
- Wink
- GazeAway
- GazeLeft
- GazeRight
- GazeUp
- GazeDown
- HeadTiltLeft
- HeadTiltRight
- HeadShake
- BigSmile
- Think
- BrowFrown